

Loan Approval Prediction

r

Presented By

Albin Manoj



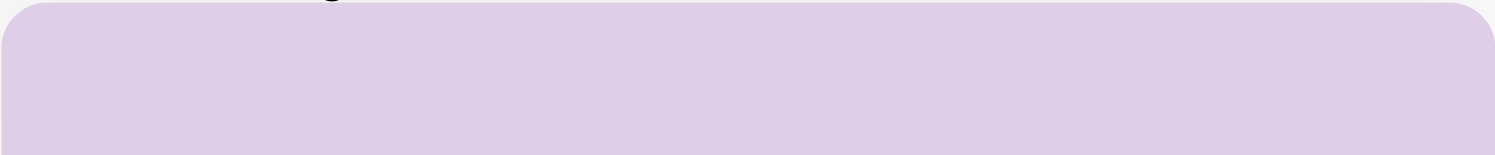


Introduction

Loan approval is a major task for financial institutions. Before approving a loan, banks must check whether the applicant is capable of repaying it. Traditionally, this process is done manually, which is slow and inefficient.

Machine Learning helps automate this process by analyzing historical applicant data and predicting loan approval status.

Banks receive many loan applications daily. Manual verification is slow, time-consuming, and may lead to mistakes. The problem is to predict whether a loan application should be approved or rejected using machine learning.



Objective

- 1 To analyze loan applicant data
- 2 To preprocess and clean the dataset
- 3 To handle missing values
- 4 To perform visualization and feature analysis
- 5 To train machine learning classification models
- 6 To predict loan approval status accurately

Dataset Description

1.Dataset Name: Loan Approval Prediction

2.Datasource: Kaggle

3.Dataset Size:

Rows:614

Columns:13

Features Used:

1.Loan_id

2.Property_area

3.Loan_status

11.Loan_amount_term

4.Gender

5.Married

6.Dependents

7.Education

8.Self_employed

9.Applicant income

10.Co applicant income

12. Credit_history

13.Loan amount

Target Variables:

Y=Loan approved

N= Not approved

Dataset

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	loan_id	gender	married	dependent	education	self_empt	applicantir	coapplicantr	loanamou	loan_amou	credit_hist	property_a	loan_status
2	lp001002	male	no	0	graduate	no	5849	0		360	1	urban	y
3	lp001003	male	yes	1	graduate	no	4583	1508	128	360	1	rural	n
4	lp001005	male	yes	0	graduate	yes	3000	0	66	360	1	urban	y
5	lp001006	male	yes	0	not gradua	no	2583	2358	120	360	1	urban	y
6	lp001008	male	no	0	graduate	no	6000	0	141	360	1	urban	y
7	lp001011	male	yes	2	graduate	yes	5417	4196	267	360	1	urban	y
8	lp001013	male	yes	0	not gradua	no	2333	1516	95	360	1	urban	y
9	lp001014	male	yes	3+	graduate	no	3036	2504	158	360	0	semiurban	n
10	lp001018	male	yes	2	graduate	no	4006	1526	168	360	1	urban	y
11	lp001020	male	yes	1	graduate	no	12841	10968	349	360	1	semiurban	n
12	lp001024	male	yes	2	graduate	no	3200	700	70	360	1	urban	y
13	lp001027	male	yes	2	graduate		2500	1840	109	360	1	urban	y
14	lp001028	male	yes	2	graduate	no	3073	8106	200	360	1	urban	y
15	lp001029	male	no	0	graduate	no	1853	2840	114	360	1	rural	n
16	lp001030	male	yes	2	graduate	no	1299	1086	17	120	1	urban	y
17	lp001032	male	no	0	graduate	no	4950	0	125	360	1	urban	y
18	lp001034	male	no	1	not gradua	no	3596	0	100	240		urban	y
19	lp001036	female	no	0	graduate	no	3510	0	76	360	0	urban	n
20	lp001038	male	yes	0	not gradua	no	4887	0	133	360	1	rural	n
21	lp001041	male	yes	0	graduate		2600	3500	115		1	urban	y
22	lp001043	male	yes	0	not gradua	no	7660	0	104	360	0	urban	n
23	lp001046	male	yes	1	graduate	no	5955	5625	315	360	1	urban	y
24	lp001047	male	yes	0	not gradua	no	2600	1911	116	360	0	semiurban	n
25	lp001050		yes	2	not gradua	no	3365	1917	112	360	0	rural	n
26	lp001052	male	yes	1	graduate		3717	2925	151	360		semiurban	n
27	lp001066	male	yes	0	graduate	yes	9560	0	191	360	1	semiurban	y
28	lp001068	male	yes	0	graduate	no	2799	2253	122	360	1	semiurban	y

Methodology

Technologies Used:

Programming language: Python

Used Platform: Google Colab

Libraries Used: Pandas, Numpy, Matplotlib, Seaborn, Scikit-learn

Machine learning Algorithms: Decision Tree, Logistic Regression, Random Forest

Visualization Techniques: Bar Chart, Histogram, Heatmap





Workflow

1. **Load dataset**
 2. **Understand the data**
 3. **Check missing values**
 4. **Perform data cleaning**
 5. **Visualize the data**
 6. **Convert categorical data to numeric**
 7. **Split dataset into training and testing sets**
 8. **Train machine learning models**
 9. **Evaluate model performance**
 10. **Compare results**
- 

Data Preprocessing

- 1.Dataset was loaded using Pandas
- 2.Checked dataset values,shape,info and missing values
- 3.Removed duplicate values
- 4.Converted Categorical data to numerical data using Label encoding
- 5.Visualized the data for more insights
- 6.Removed Unnecessary Columns

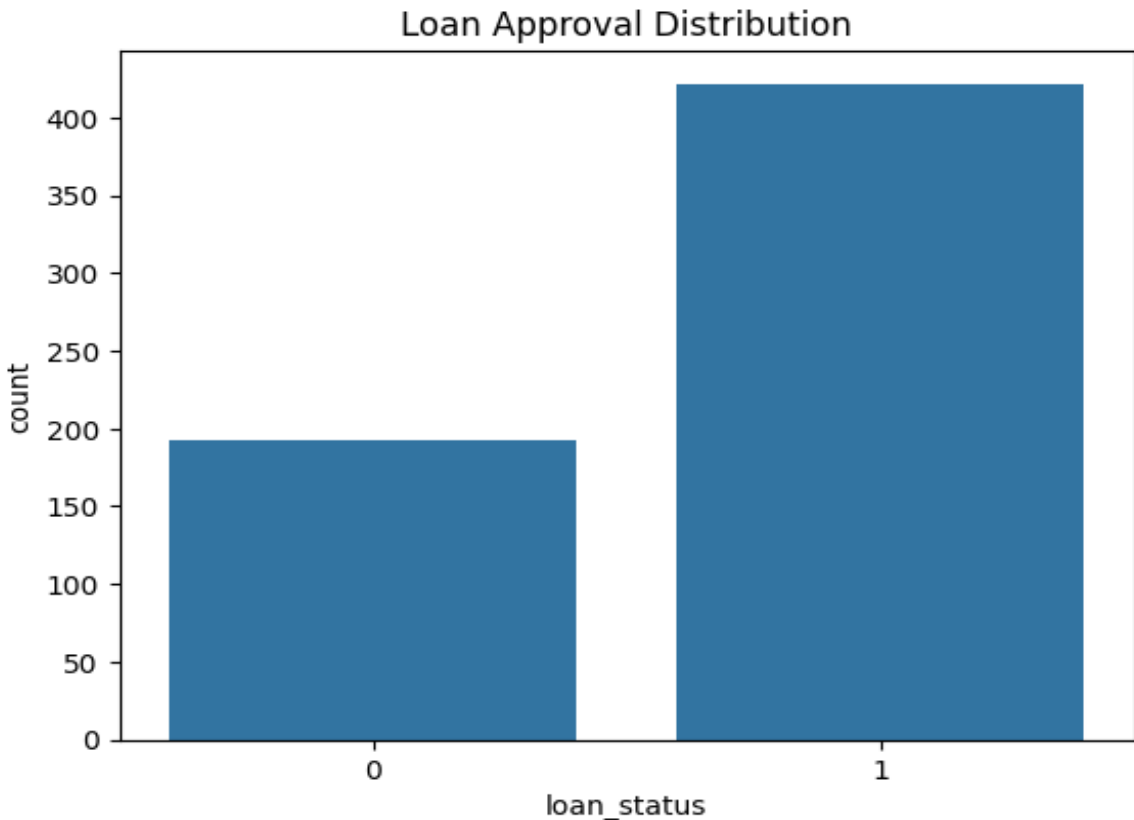
Exploratory Data Analysis

- .Understand Data Structures
- .Find Missing Values
- .Understand Data Distribution
- .Detect relationship between features
- .Improve model accuracy

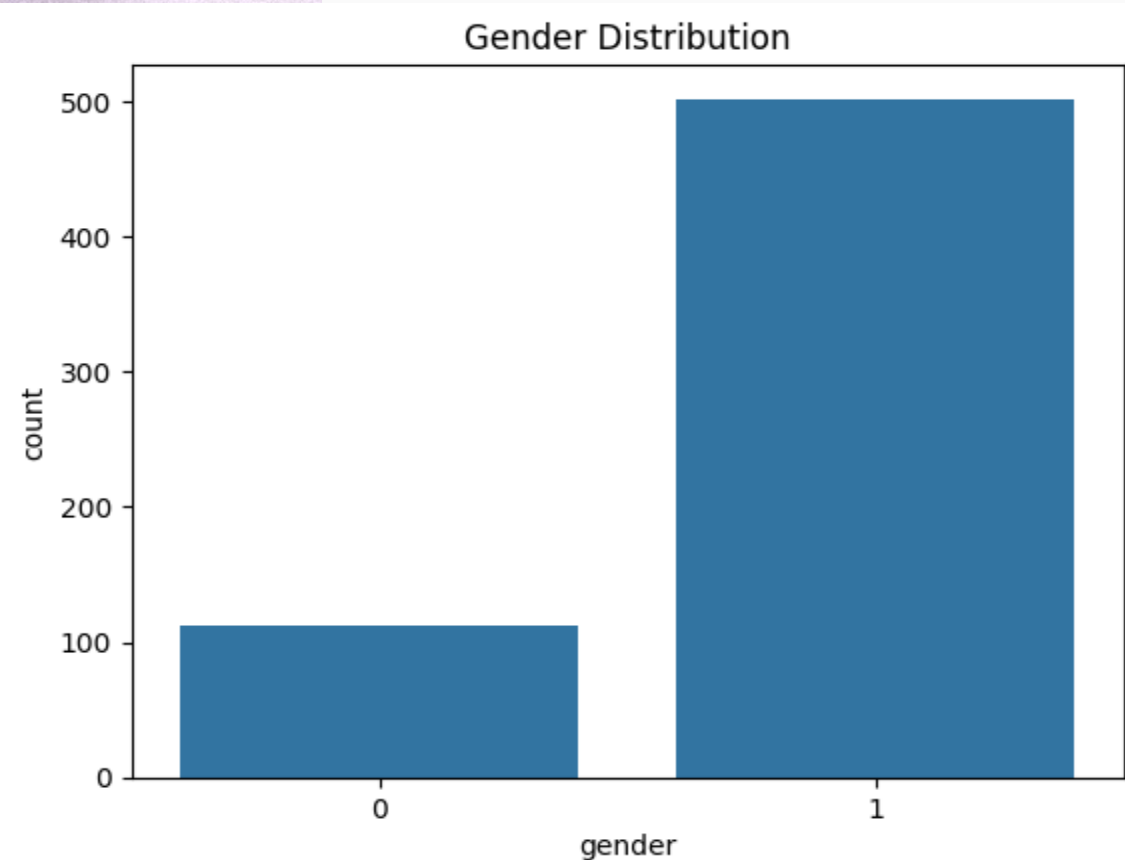
Loan Approval Distribution

Insights:

- 1.The number of approved loans (**1**) is higher than rejected loans (**0**).
- 2.This shows that most applicants in the dataset were eligible for loan approval.
- 3.It also indicates that the dataset is slightly imbalanced toward approved loans.



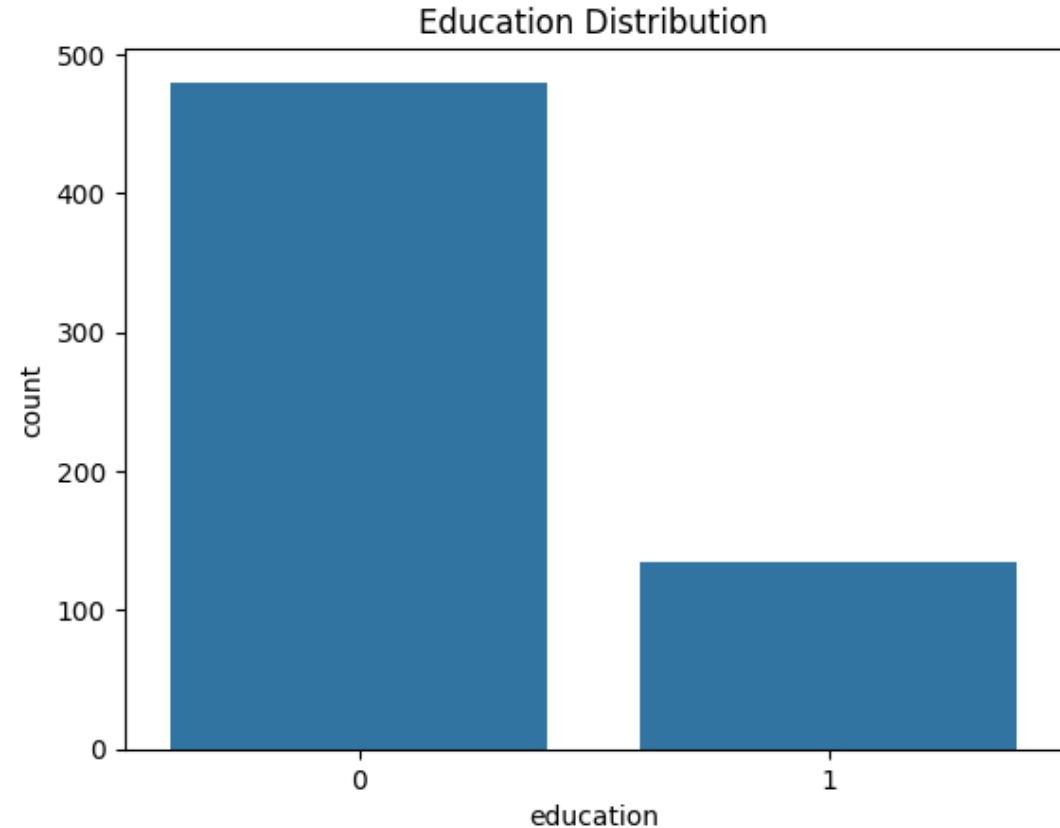
Gender Distribution



Insights:

1. The number of male applicants is much higher than female applicants.
2. This indicates that more male applicants applied for loans compared to female applicants in this dataset.

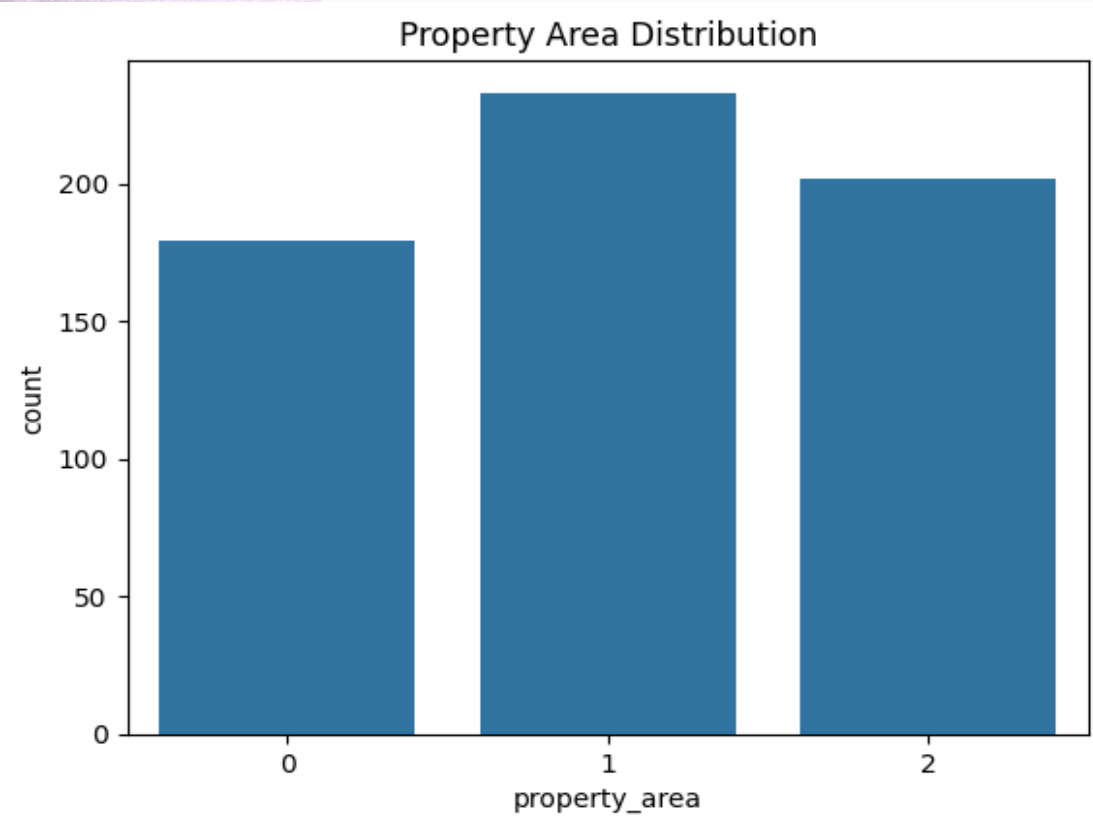
Education Distribution



Insights:

1. Graduate applicants are more than non-graduate applicants.
2. This suggests that educated applicants are more likely to apply for loans.
3. Education may also influence loan approval decisions.

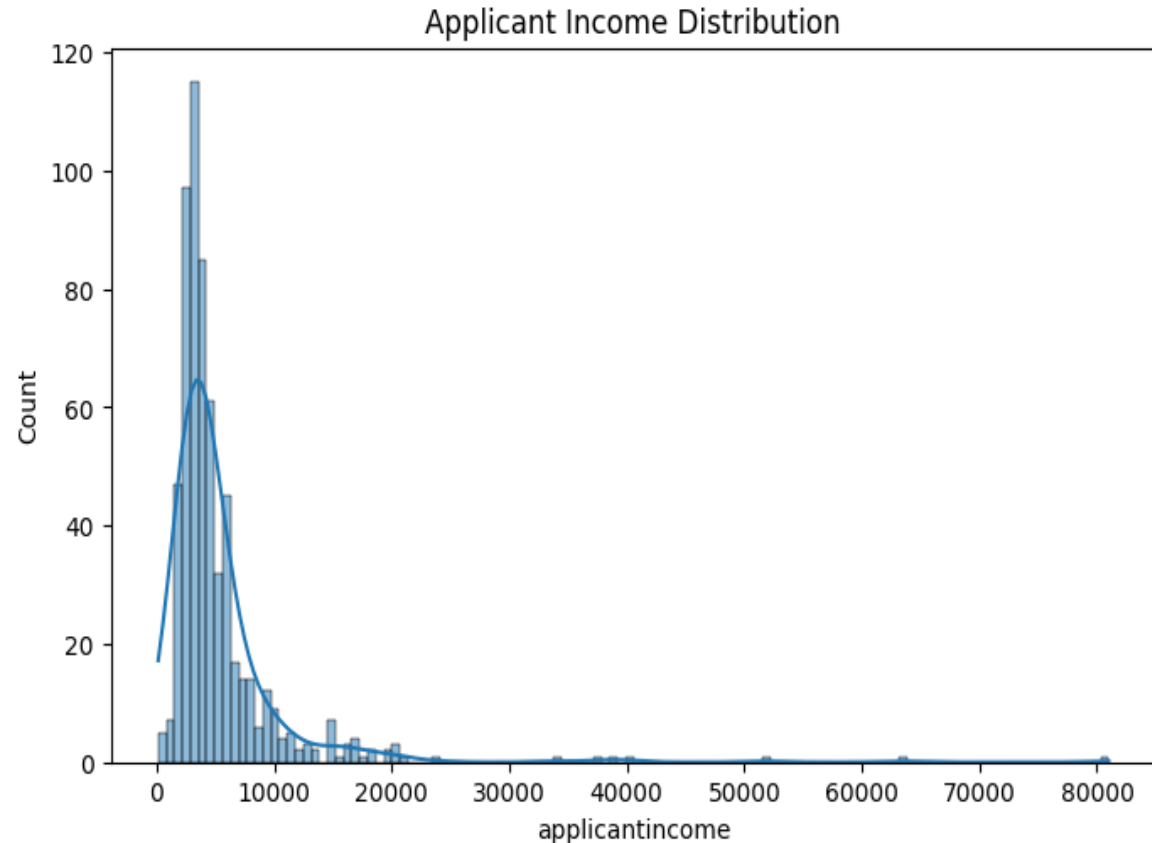
Property Area Distribution



Insights:

1. Most applicants belong to Semi urban areas, followed by Urban and Rural areas.
2. This shows that semiurban applicants are the most common in the dataset.
3. Location may affect loan approval patterns.

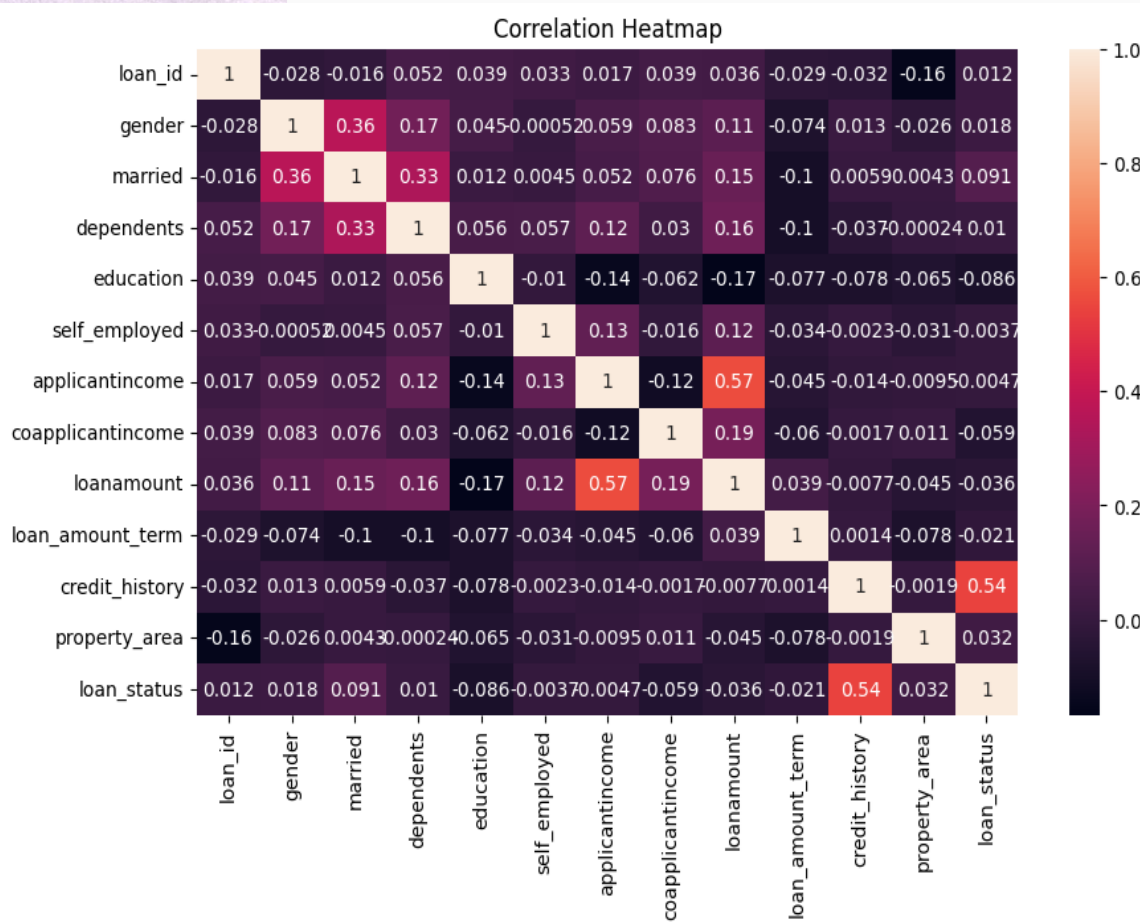
Applicant Income Distribution



Insights:

1. Most applicants have income in the lower to middle income range.
2. Only a few applicants have very high income, creating a right-skewed distribution.
3. This means income is not evenly distributed.

Correlation Heatmap



Insights:

1. Credit history shows a strong relationship with loan approval.

2. Applicant income and loan amount also show some relationship.

3. This indicates that credit history is one of the most important factors for loan approval prediction.

Data Splitting

The dataset was divided into:

- Training Data (80%)
- Testing Data (20%)

Evaluates Training Model
for better performance

Trains the data for model
training

Algorithms Used

Logistic Regression

A basic classification algorithm used for binary prediction.

Decision Tree

A tree-based model used for decision-making and classification.

Random Forest

An advanced ensemble model that combines multiple decision trees for better accuracy.

Model Evaluation

Evaluation Metrics

The following metrics were used:

- Accuracy Score
- Confusion Matrix
- Classification Report

Accuracy

.The models accuracy is 78%:

.Random Forest performed better than simpler models and gave reliable results.

Confusion Matrix:	
[[18 25]	
[2 78]]	
Actual Result	Predicted Result
Rejected Loan	Count
Correctly Rejected	18
Rejected Loan	
Wrongly Approved	25
Approved Loan	
Wrongly Rejected	2
Approved Loan	
Correctly Approved	78

Model Evaluation

Observation:

The model performs very well in predicting approved loans but is slightly weaker in predicting rejected loans.

Classification Report

Important Values:

For Approved Loans (Class 1)

- Precision = 0.76
- Recall = 0.97
- F1-score = 0.85

Observation:

The recall value is very high, which means the model successfully identifies most approved loan applications.

For Rejected Loans (Class 0)

- Precision = 0.90
- Recall = 0.42
- F1-score = 0.57

Observation:

The recall is lower, meaning the model misses some rejected loan applications.



Hyper Parametric Tunning

Objective

To optimize the Random Forest model by finding the best parameter values and improving model performance.

Technique Used

.GridSearchCV with 5-Fold Cross Validation

Parametric Testing

.Hyper parameter	Value tested
-------------------------	---------------------

n_estimators	50,100,200
--------------	------------

Max_depth	5,10,none
-----------	-----------

Best value of n_estimator is 100 and max_depth is 5

**The improved Accuracy after hyper parametric tuning is 81%.
It has increased accuracy of 5% than the previous accurate value**

App Deployment

- .We have created an app for Loan Approval Prediction.
- .We created the app using Streamlit.
- .There were two files for the app where one file had the code for the app and another file had requirements.
- .Libraries used for the app
 - 1.Streamlit
 - 2.Pandas
 - 3.Pickle
- .Libraries used for the requirement
 - 1.Pandas
 - 2.Streamlit
 - 3.Numpy
 - 4.Scikit-learn

Results

.Among all models, Random Forest gave better prediction accuracy because it handles multiple features efficiently and reduces overfitting.

.Credit History was found to be one of the most important features affecting loan approval.

Thus the project successfully predicts whether a loan should be approved or rejected.





Thank you

